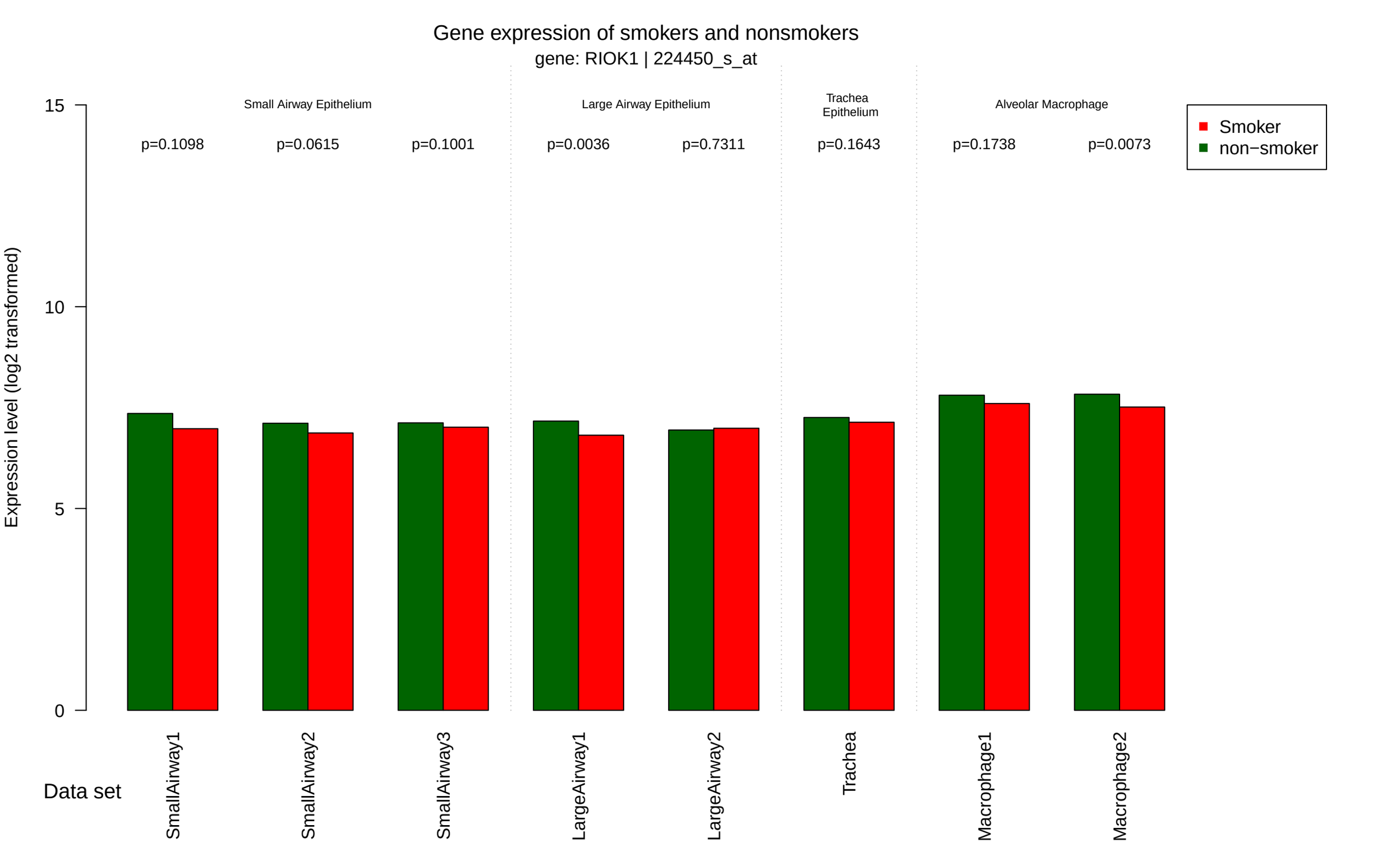
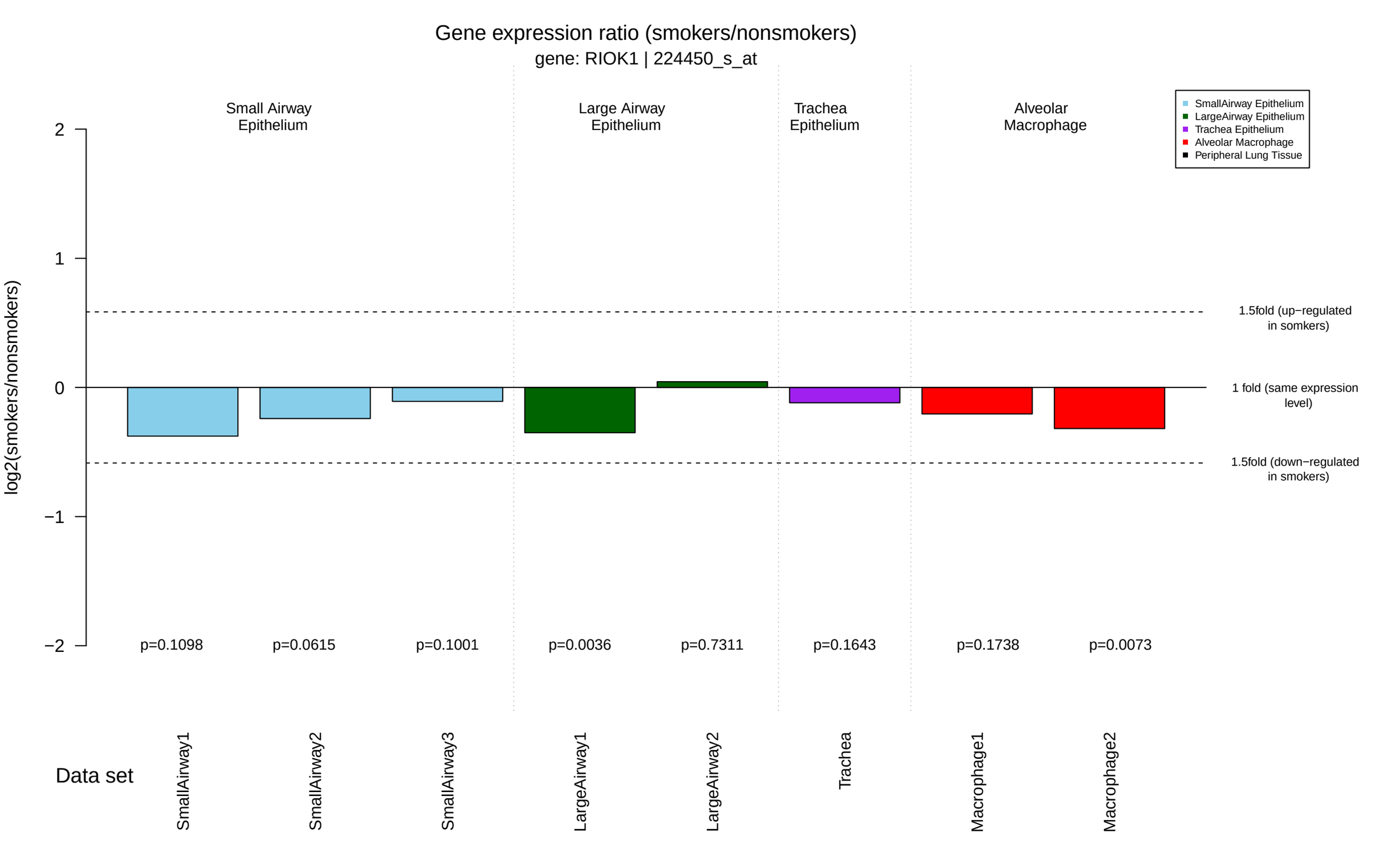
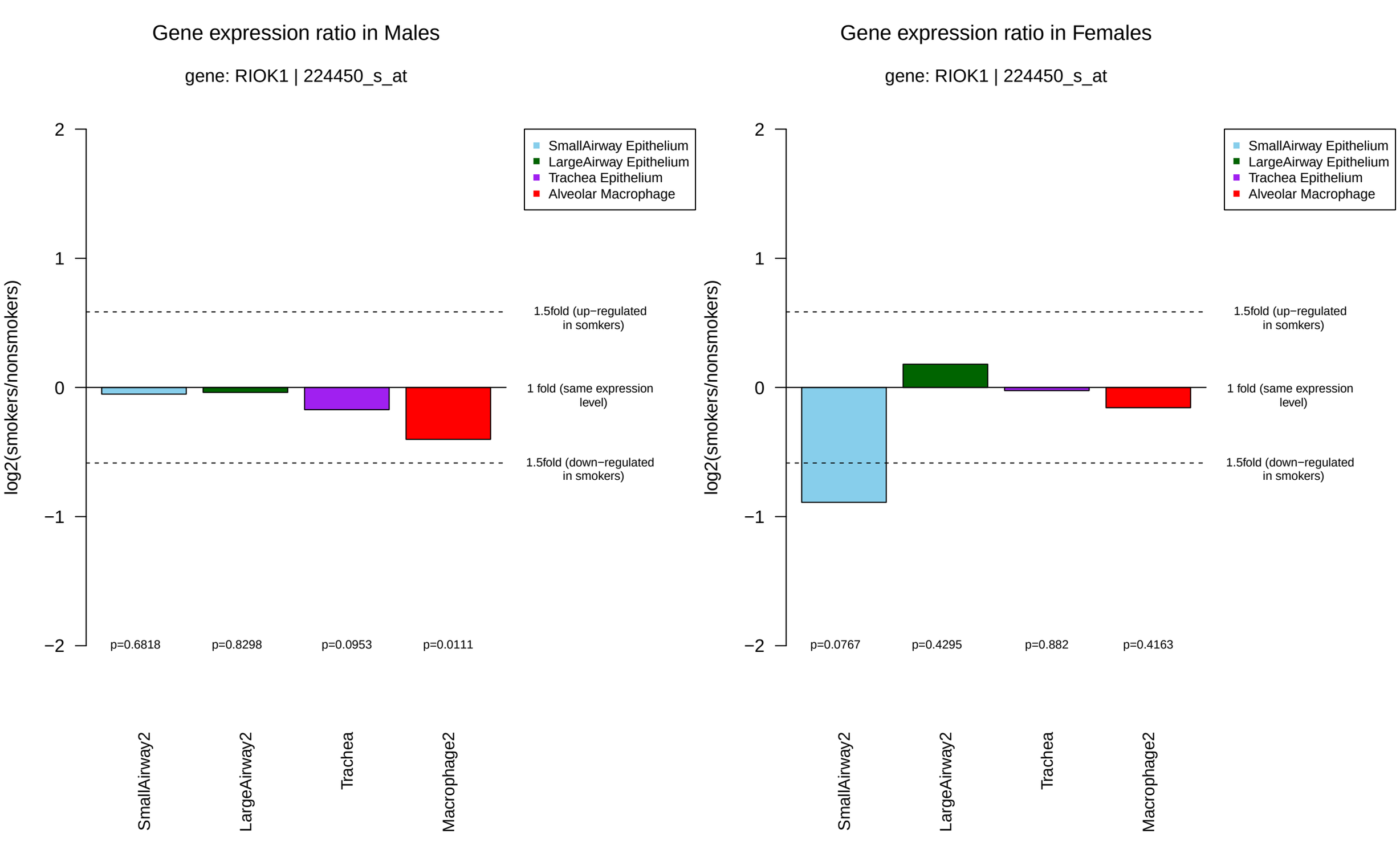
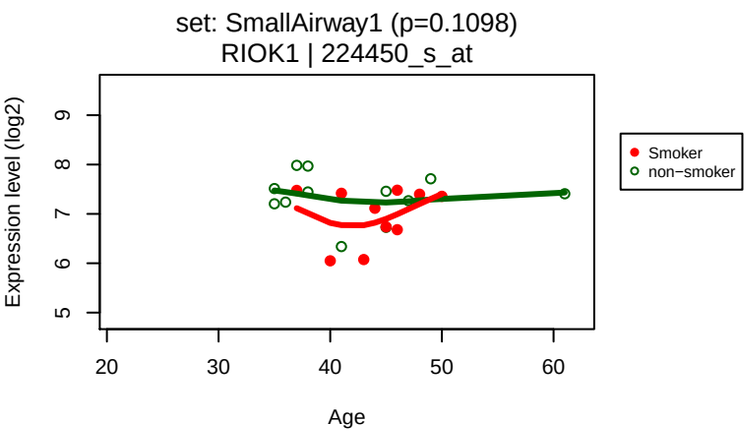
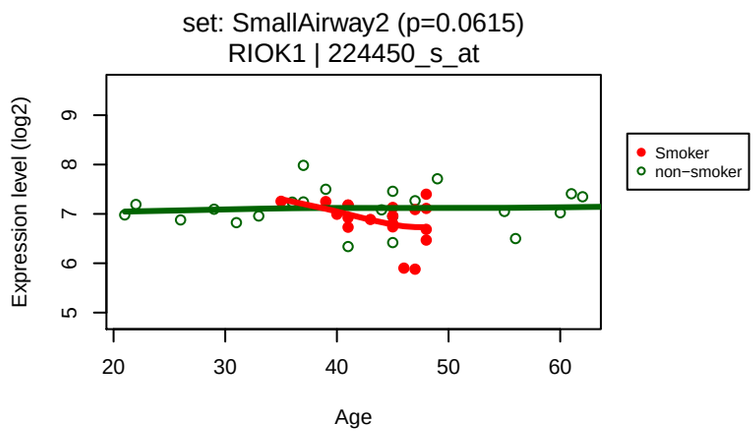
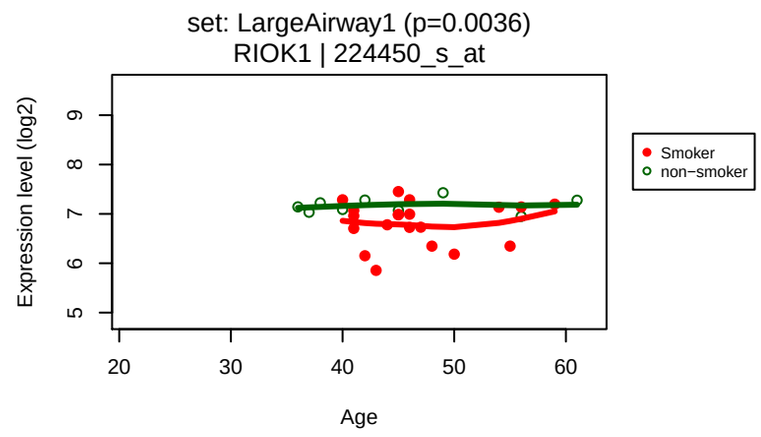

Gene expression of smokers and nonsmokers
gene: RIOK1 | 224450_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1098
p=0.0615
p=0.1001
p=0.0036
p=0.7311
p=0.1643
p=0.1738
p=0.0073
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

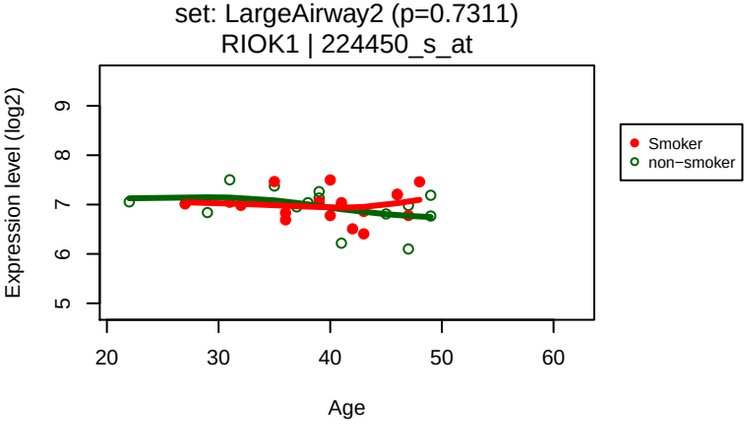

Gene expression ratio (smokers/nonsmokers)
gene: RIOK1 | 224450_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1098
p=0.0615
p=0.1001
p=0.0036
p=0.7311
p=0.1643
p=0.1738
p=0.0073
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

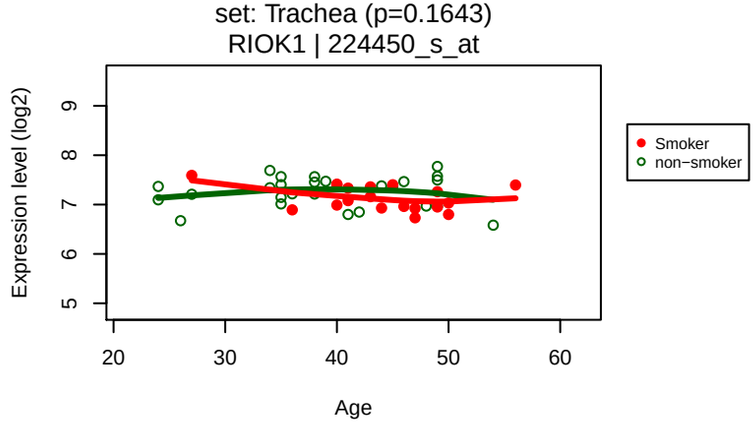

Gene expression ratio in Males
gene: RIOK1 | 224450_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6818
p=0.8298
p=0.0953
p=0.0111
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RIOK1 | 224450_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0767
p=0.4295
p=0.882
p=0.4163
SmallAirway2
LargeAirway2
Trachea
Macrophage2

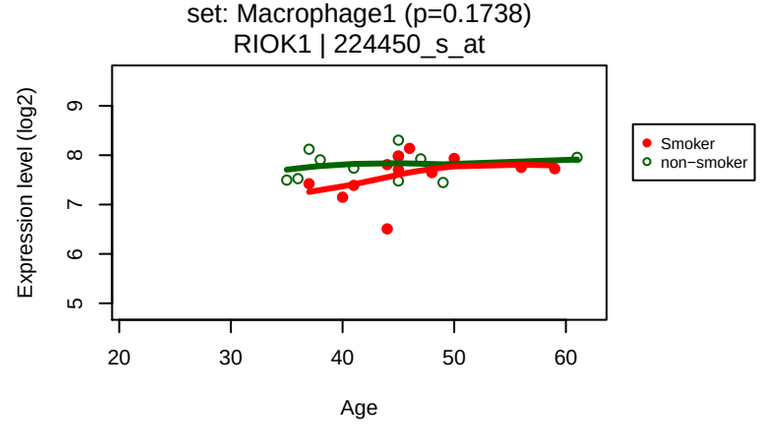

set: SmallAirway1 (p=0.1098)
RIOK1 | 224450_s_at
Expression level (log2)
Age
Smoker
non−smoker

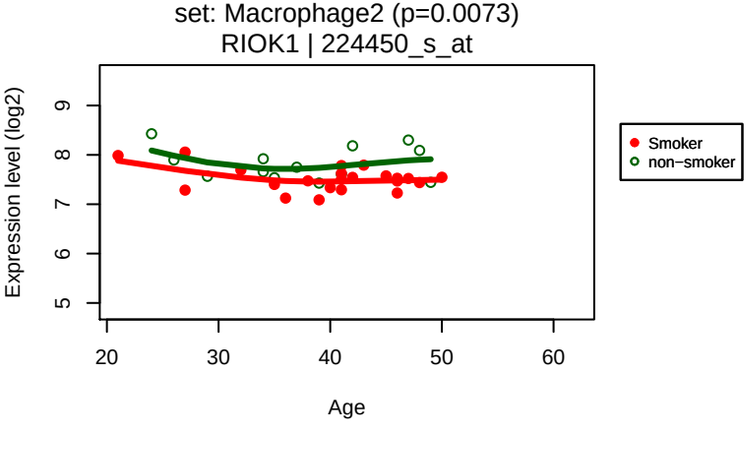

set: SmallAirway2 (p=0.0615)
RIOK1 | 224450_s_at
Expression level (log2)
Age
Smoker
non−smoker

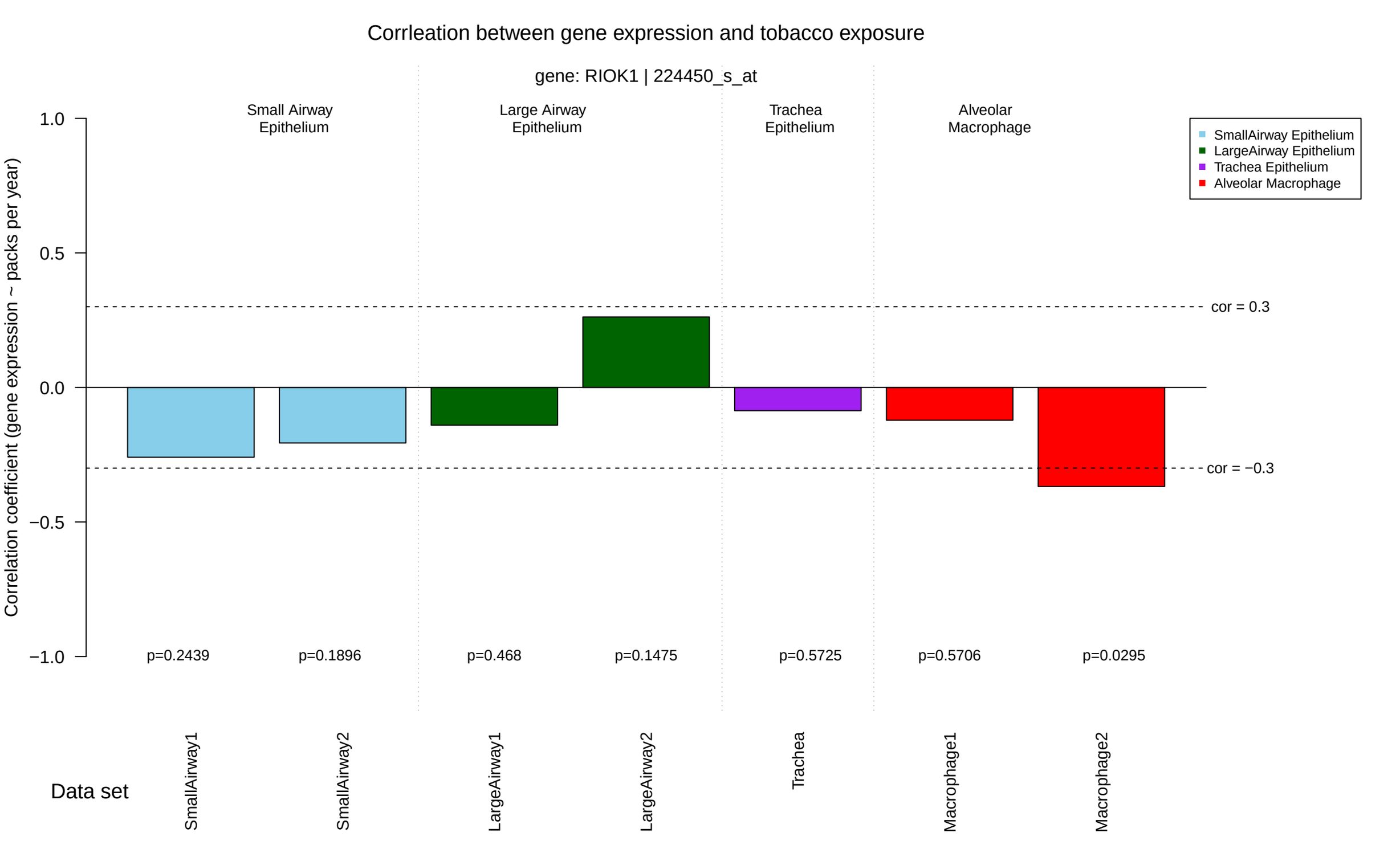

set: LargeAirway1 (p=0.0036)
RIOK1 | 224450_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7311)
RIOK1 | 224450_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1643)
RIOK1 | 224450_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1738)
RIOK1 | 224450_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0073)
RIOK1 | 224450_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RIOK1 | 224450_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2439
p=0.1896
p=0.468
p=0.1475
p=0.5725
p=0.5706
p=0.0295
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2